2026 prelim waves questions

Wave questions from the same 2026 prelim pool as the mechanics and electricity sets.
Oscillation-only questions are not included.
Short questions whose only content is superposition are not included.
Eleven questions remained, so no papers outside this pool were added.
Whole pages are kept. A page that also holds another question is included in full.

2026 RI P2 Q2 p.2
2026 RI P3 Q4 p.3
2026 DHS P2 Q3 p.6
2026 EJC P2 Q5 p.11
2026 EJC P3 Q4 p.13
2026 EJC P3 Q6 p.15
2026 HCI P2 Q3 p.17
2026 HCI P2 Q4 p.19
2026 HCI P3 Q10 p.21
2026 NYJC P2 Q4 p.27
2026 VJC P3 Q3 p.30

2 **(a)** State what is meant by *coherent* light waves.

.....

..... [1]

(b) Coherent light of wavelength 530 nm is incident normally on a double slit, as illustrated in Fig. 2.1.

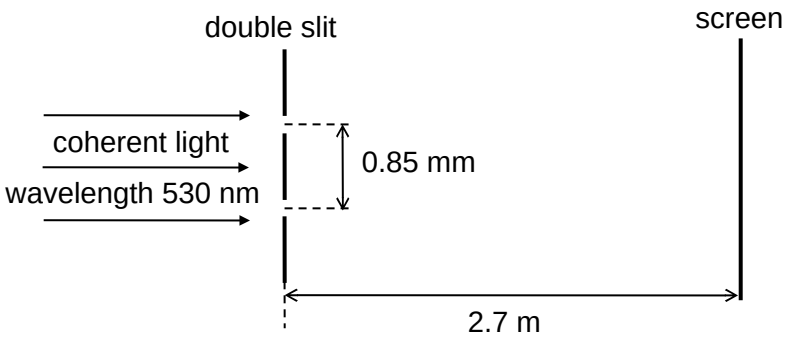


Fig. 2.1 (not to scale)

The separation of the two slits is 0.85 mm. A screen is placed parallel to the plane of the double slit at a distance of 2.7 m from the double slit.

Determine the separation of the fringes observed on the screen.

separation = ... m [2]

(c) Describe the changes to the appearance of the fringes when each of the following changes is made separately.

(i) The width of both slits is reduced without altering their separation.

.....

..... [1]

(ii) The screen is rotated so that it is no longer parallel to the plane of the two slits.

.....

.....

.....

.....

..... [2]

[Total: 6]

4 **(a)** An incoming polarised beam of monochromatic light from a light source passes through two plane polarisers. I_0 and I_1 are the intensities of the light along the central axis just before and after passing through the second polariser respectively as shown in Fig. 4.1.

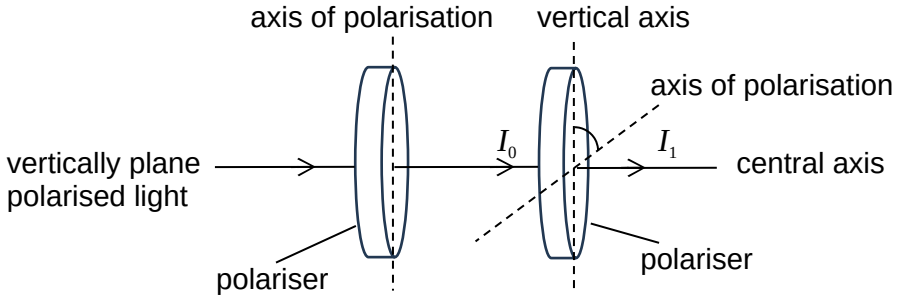


Fig. 4.1

The incoming light is polarised in the vertical plane. The two polarisers have their axis of polarisation at an angle θ to one another and the variation of I_1 with $\cos^2\theta$ is shown in Fig. 4.2.

I_1 / W m^{-2}

Fig. 4.2

(i) Determine two possible values of θ such that $\frac{I_1}{I_0} = 0.50$ where $0^\circ < \theta < 360^\circ$.

$\theta =$ ..... ° and ..... ° [2]

(ii) The light source is adjusted such that the intensity of the light just before passing through the second polariser is now $\frac{1}{3} I_0$.

Draw a line in Fig. 4.2 to indicate the new variation of the intensity I_1 with $\cos^2\theta$.

[1]

(b) The polarisers are now removed and a light intensity detector is placed at different horizontal distances from the light source which is a point source of light.

The detector indicates a value of 0.20 W m^{-2} when it is at a distance of 20 cm from the light source.

On the axes of Fig. 4.3, sketch a graph to show the variation of the intensity I_d measured by the detector with distance r from the source. The graph should extend from $r = 20$ cm to $r = 60$ cm.

Include an appropriate scale for the intensity axis.

I_d / W m^{-2}

Fig 4.3

[3]

[Total: 6]

5 A capacitor of capacitance *C* is connected in parallel with a resistor of resistance 200 Ω across an alternating supply as shown in Fig. 5.1.

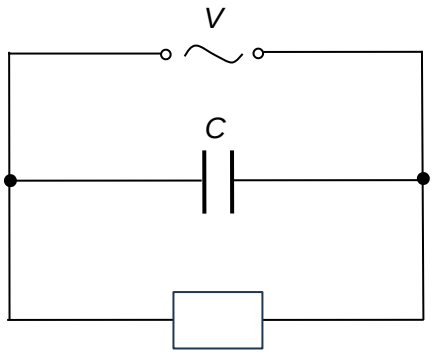


Fig. 5.1

The variation with time *t* of the potential difference *V* of the alternating supply is given by

$$V = 120\sin(314\,t),$$

where *V* is measured in volts and *t* in seconds.

(a) (i) Write an equation, in terms of *C*, that relates how charge *Q* on the capacitor plate varies with *t*.

[1]

(ii) Hence, show the current I_C through the capacitor is $37700C\cos(314\,t)$.

3 An experiment is carried out, using a light intensity meter attached to a toy train, to measure the diffraction of laser light through a double slit, as shown in Fig. 3.1 below.

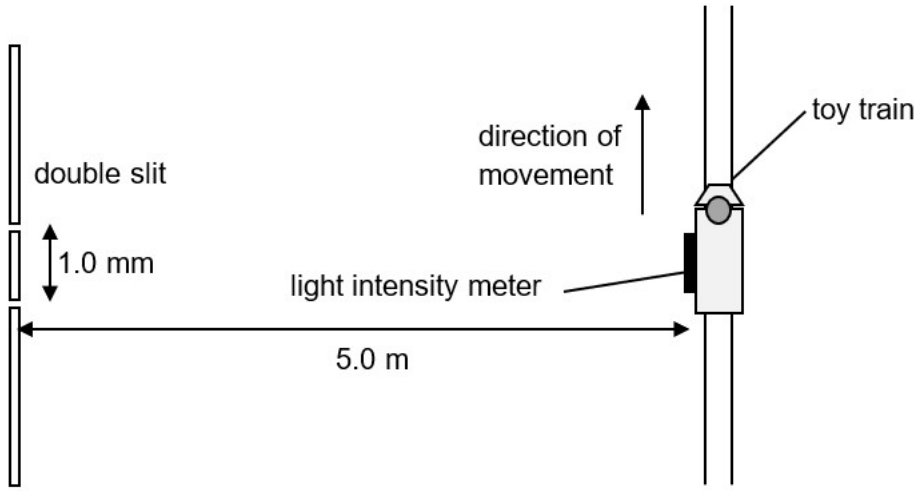


Fig. 3.1

Light from the laser is incident on the double slit. The light from the double slit is then detected by a light sensor attached to the train.

As the train moves along the track, the light intensity is recorded by the light intensity meter. The variation with time of the light intensity is shown in Fig. 3.2.

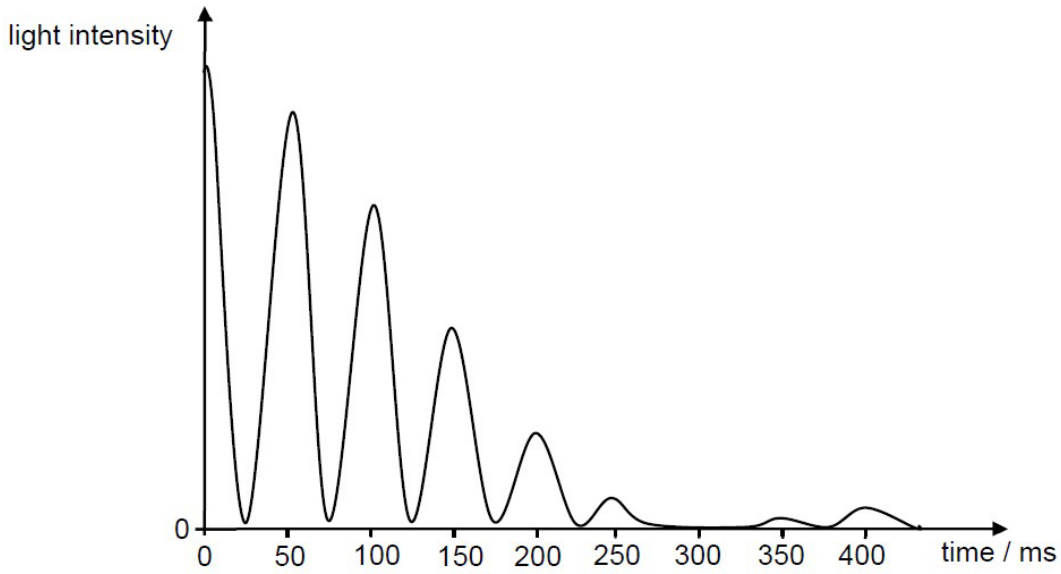


Fig. 3.2

(a) Explain, with reference to the paths travelled by the light emerging from each slit, why maxima and minima are observed.

……..

……..

……..

……..

……..

……..

……..… [2]

(b) **(i)** The speed of the train is 6.5 cm s^{-1}.

Use Fig. 3.2 to show that the fringe separation of the two slit interference pattern is 3.3×10^{-3} m.

[1]

(ii) The separation of the slits is 1.0 mm and the track is 5.0 m from the slits.

Determine the wavelength of the laser light.

wavelength = ... m [2]

(c) A maximum is detected by the light intensity meter at every 50 ms, explain why a minimum is detected at 300 ms in Fig. 3.2.

……..

……..

……..

……..

……..

……..

……..…… [1]

(d) Determine the width of one of the slits.

width = ... m [2]

(e) The experiment is changed so that the toy car is now used to investigate the formation of a stationary sound wave as shown in Fig. 3.3 below.

The light sensor of the train is replaced with a sound sensor attached to a data logger. A loudspeaker emitting sound waves of constant amplitude and frequency is placed such that it faces a reflecting barrier. The train track is adjusted so that the train is moving away from the loudspeaker towards the barrier.

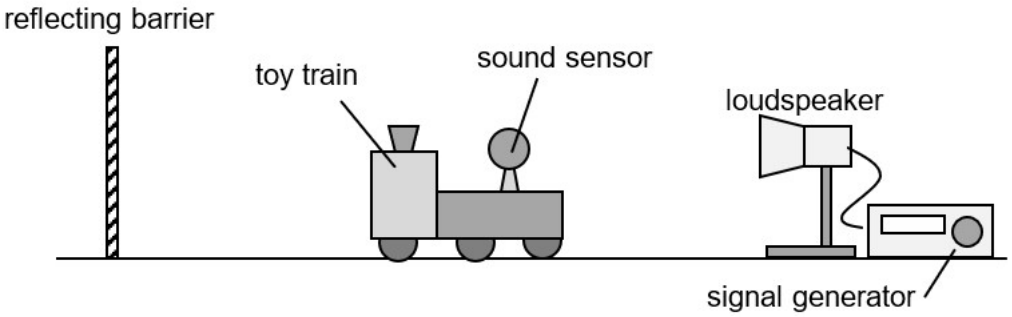


Fig. 3.3

The variation with distance of the sound intensity recorded by the sound sensor is shown in Fig. 3.4.

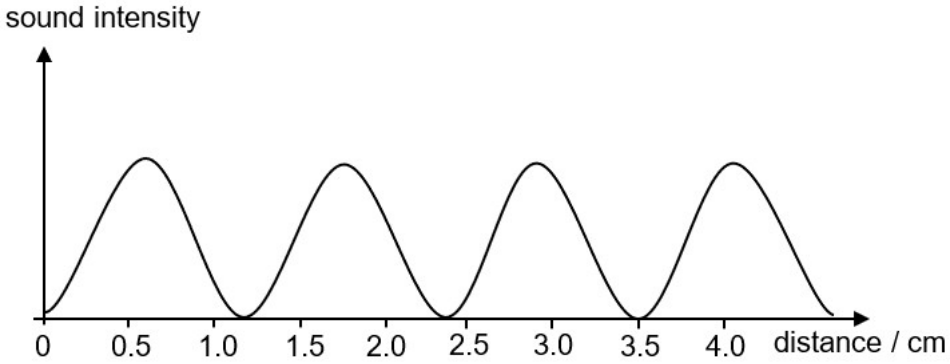


Fig. 3.4

(i) Explain how a stationary sound wave forms between the loudspeaker and reflecting barrier.

……

……

……

……

……

…… [2]

(ii) Using Fig. 3.4, determine the wavelength of the stationary wave.

wavelength = ... m [2]

(iii) Given that the frequency of the sound is 14 kHz.
Determine the speed of the sound wave.

speed = ... m s^{-1} [2]

[Total: 14]

5 A double slit consists of two parallel slits, each of width 0.12 mm. The separation of the slits is 2.8 mm, as illustrated in Fig. 5.1.

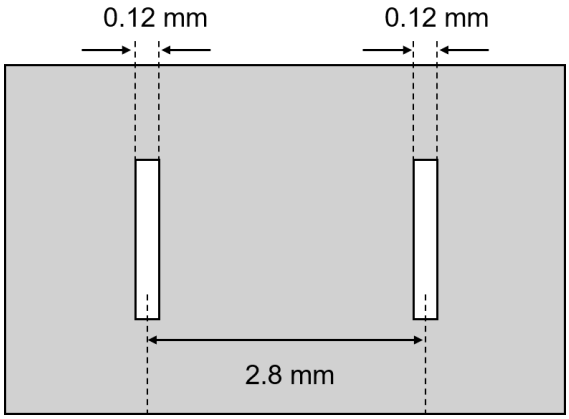


Fig. 5.1 (not to scale)

Parallel light of wavelength 670 nm is incident normally on the double slit.

A screen is placed parallel to the plane of the double slit at a distance of 1.2 m from the slits, as illustrated in Fig. 5.2

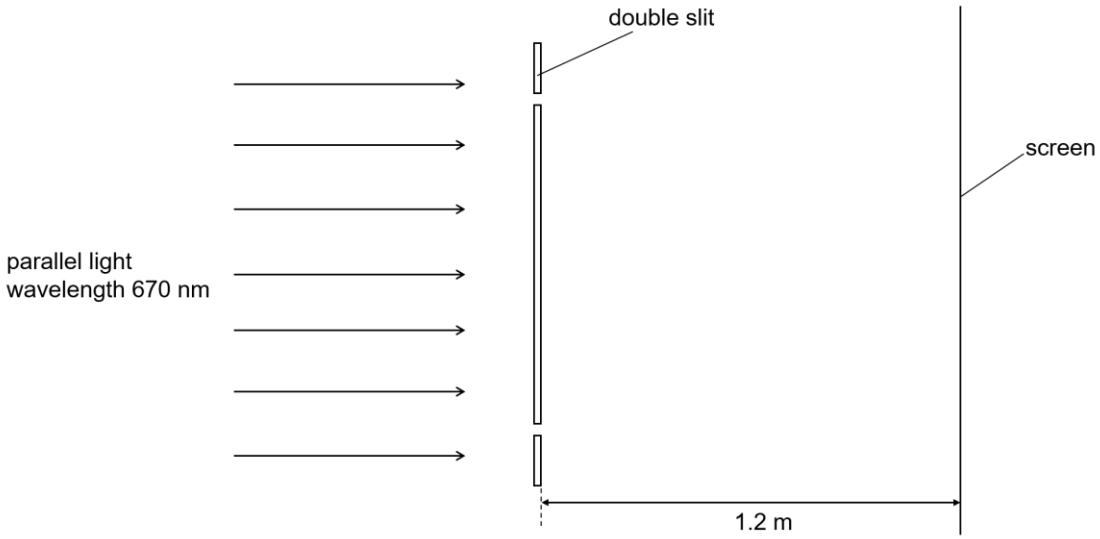


Fig. 5.2 (not to scale)

(a) Initially, one of the two slits is covered.

Show that the width of the central fringe for a single-slit diffraction pattern on the screen is 13.4 mm.

[3]

(b) Both slits are now uncovered.

(i) Use the Rayleigh criterion to explain whether the diffraction patterns produced by the two slits are seen on the screen as being separated.

……

……

……

……

…… [2]

(ii) Using your answer in **(a)**, estimate the number of fringes within the central bright maximum.

number = …….. [2]

[Total: 7]

4 **(a)** A radio mast transmits a signal using a transmitter with output power 15.0 W. The signal is received at a research station 3.5 km away.

(i) Calculate the power received by a receiving antenna of effective area 450 cm².

power received = ……W [2]

(ii) State the assumption made in your calculation of part (a)(i).

……[1]

(iii) The amplitude of the signal at a distance of 2.5 km from the transmitter is A_0. Determine, in terms of A_0, the amplitude of the signal received at the antenna at a distance of 3.5 km.

amplitude = …… A_0 [2]

(b) **(i)** State what is meant by a polarised wave.

……

……[1]

(ii) A vertically polarised light of intensity I_o is incident on a polariser P with its transmission axis at an angle of 40° from the vertical as shown in Fig. 4.1. The wave then passes through another polariser Q with its transmission axis at an angle of 40° from the vertical in the other direction.

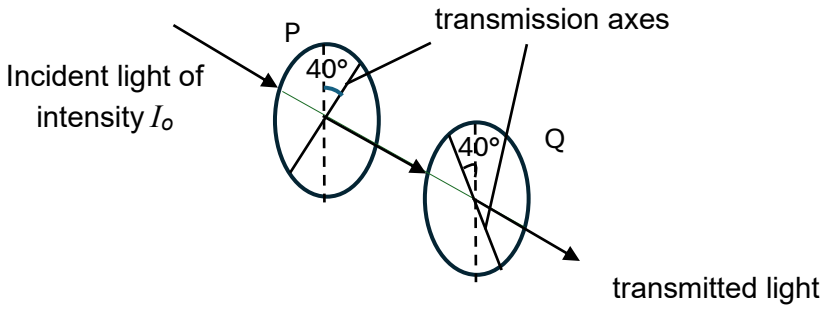


Fig. 4.1

Determine the intensity of the transmitted light in terms of I_o.

intensity = …… I_o [2]

[Total: 8]

6 **(a)** State the principle of superposition.

……

…….. [1]

(b) A 15.0 m long glass tube has a loudspeaker, connected to a signal generator, placed at one end.

A microphone connected to a cathode-ray oscilloscope (CRO) can be moved through the tube as shown in Fig. 6.1.

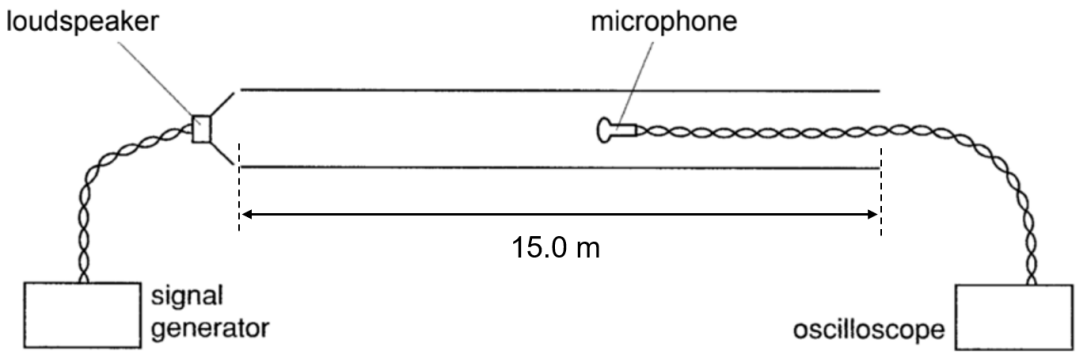


Fig. 6.1

State the conditions for the formation of a stationary wave and explain how an observable stationary wave is formed in the long glass tube.

……

……

……

……

……

……..... [3]

(c) The speaker produces sound waves with speed 320 m s^{-1} and an amplitude of 0.060 mm.

The frequency of the wave is adjusted until a loud sound is heard.

(i) State the maximum amplitude of the stationary wave

maximum amplitude = ... mm [1]

(ii) The microphone is moved from one end to the other within the 15.0 m long glass tube and detects a total of three pressure antinodes.

Complete Fig. 6.2 to represent the stationary wave in the glass tube.

Fig. 6.2

[2]

(iii) Using your answer in **(c)(ii)**, determine the frequency of the stationary wave.

frequency = ... Hz [1]

[Total: 8]

3 A beam of unpolarised light of intensity I_o is incident normally on a set-up consisting of three polarisers P_1, P_2 and P_3, arranged as shown in Figure 3.1.

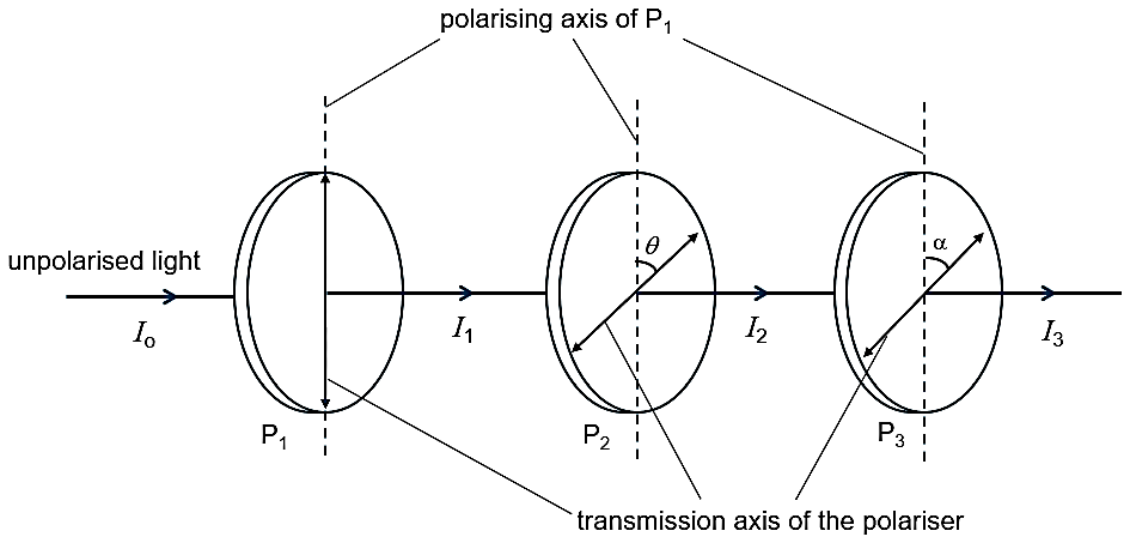


Figure 3.1

The light is plane polarised vertically after passing through P_1. The transmission axis of P_2 is at an angle θ to the vertical and the transmission axis of P_3 is at an angle α to the vertical.

The intensity of the light transmitted through P_1, P_2 and P_3 are I_1, I_2 and I_3 respectively.

(a) Calculate the transmitted intensity I_3 in terms of I_o when $\theta = 25°$ and $\alpha = 90°$.

I_3 = ……..…….. [2]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

(b) Angle α of polariser P_3 is kept fixed at 90° while angle θ of polariser P_2 is varied.

In Figure 3.2, sketch the variation of the amplitude of transmitted light after passing through P_3 with θ. Numerical values on the vertical axis are not needed.

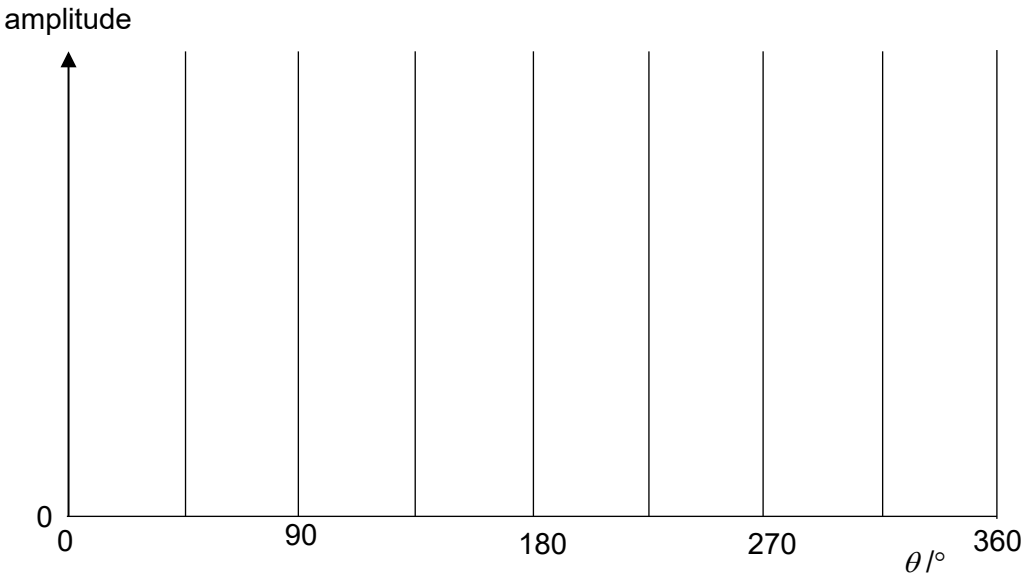


Figure 3.2

[2]

[Total: 4]

4 A receiver aerial is placed between a point source emitting microwaves at a single fixed wavelength of 140 mm and a reflective metal sheet, as shown in Figure 4.1.

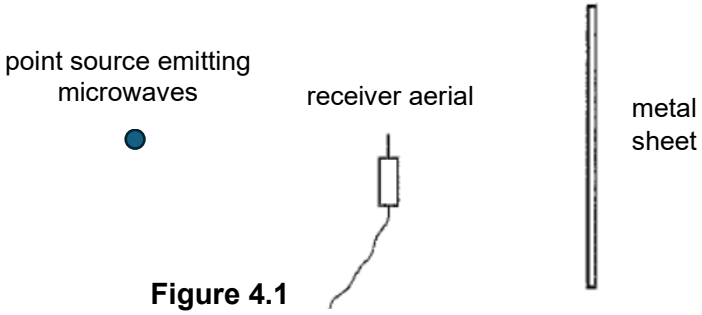


Figure 4.1

The receiver aerial is fixed in position. The metal sheet is moved away from the receiver aerial toward the right. The detected voltage at the receiver aerial varies with the distance moved by the sheet as shown in Figure 4.2.

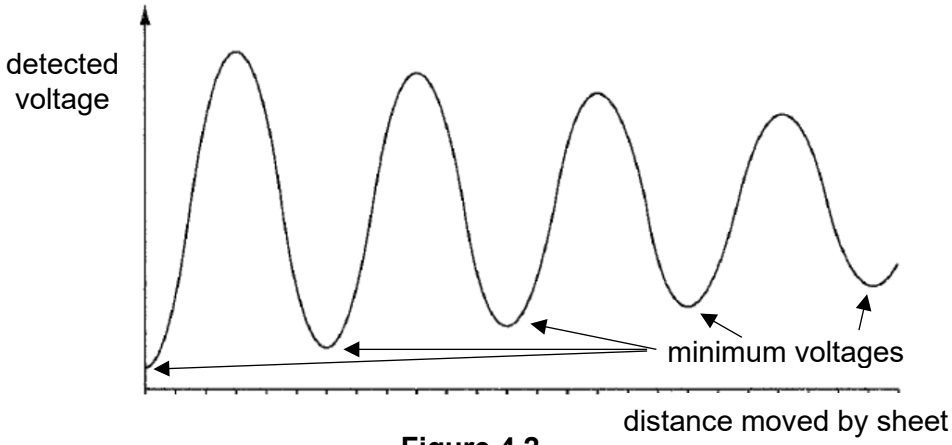


Figure 4.2

(a) Explain why the minimum voltage detected at the receiver aerial gets larger with distance moved by the sheet.

.....

.....

.....

.....

.....

.....

.....

..... [3]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

(b) State the distance moved by the metal plate between consecutive detection by the aerial receiver of minimum voltages.

distance = ... mm [1]

[Total: 4]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

10 **(a)** State what is meant by coherent light waves.

.....

..... [1]

(b) Coherent light of wavelength of 633 nm is incident normally on a pair of narrow parallel slits separated by a distance of 0.250 mm as shown in Figure 10.1.

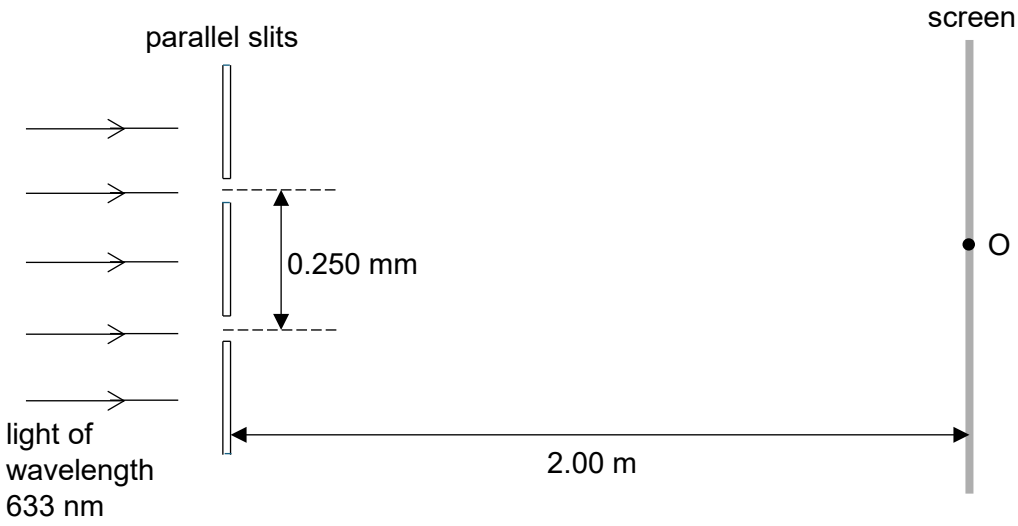


Figure 10.1 (not to scale)

An interference pattern of bright and dark fringes is seen on a screen placed parallel to the plane of the slits. The screen is 2.00 m away from the slits. Point O is equidistant from both slits.

(i) Explain how the bright and dark fringes are formed on the screen.

.....

.....

.....

.....

..... [2]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

(ii) Calculate the distance between the central bright fringe and the second-order dark fringe on the screen.

distance = ... mm [2]

(iii) Explain why the maximum observed at O is of a greater intensity than any other observable maximum.

.....

.....

.....

.....

..... [2]

(c) The setup is modified by blocking one of the slits, leaving a single slit of width 0.080 mm exposed to the original light source of single wavelength of 633 nm as shown in Figure 10.2.

A diffraction pattern of the single slit is formed on the screen centred at point P.

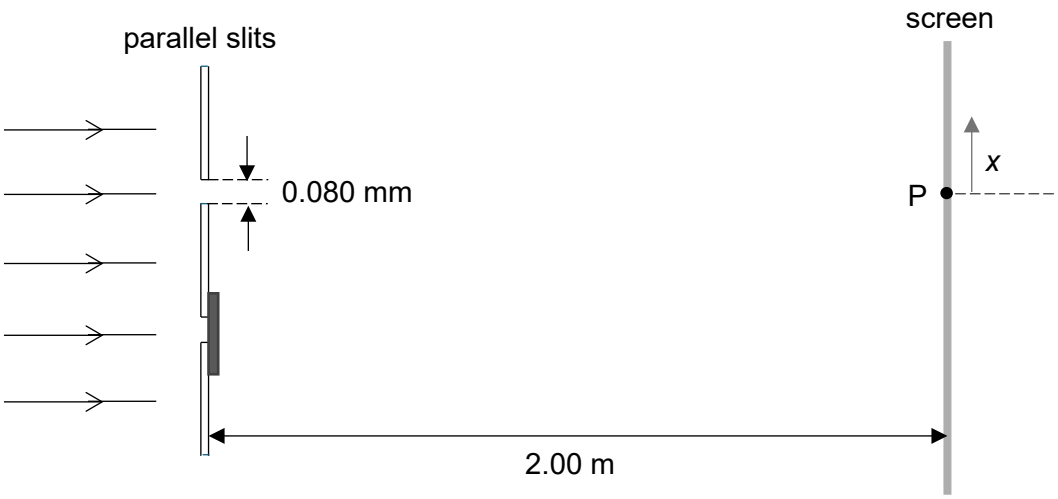


Figure 10.2 (not to scale)

(i) Show that the width of the central fringe on the screen is approximately 31.7 mm.

[2]

(ii) Part of the diffraction pattern showing the variation with displacement x from P of the intensity of light is shown in Figure 10.3.

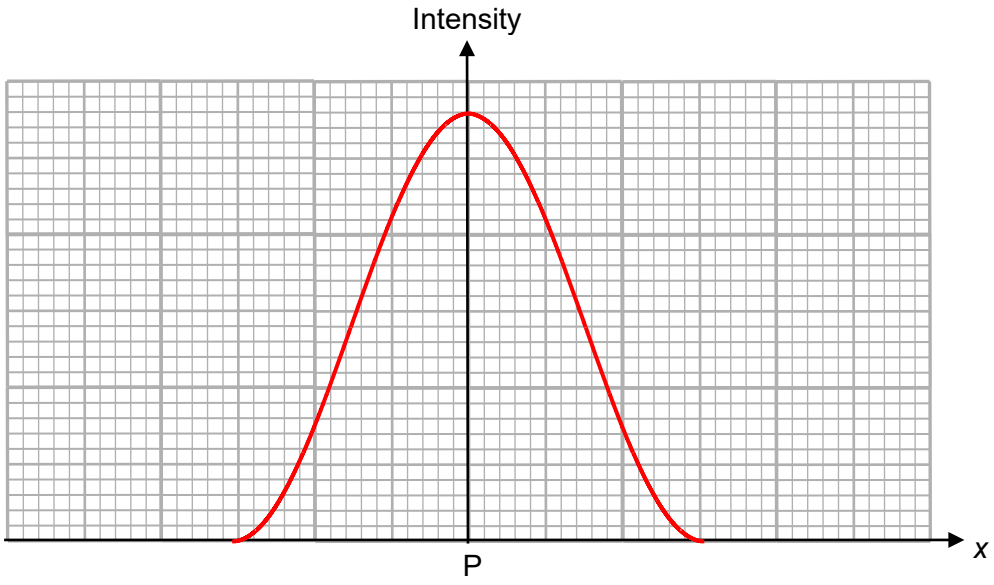


Figure 10.3

The slit width is reduced by half. Sketch on Figure 10.3 the new diffraction pattern. [2]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

(iii) The single slit is now used as an aperture for a simple optical imaging system.

Two small LED indicators are spaced 5.0 mm apart as shown in Figure 10.4. The LED indicators emit light of wavelength 633 nm.

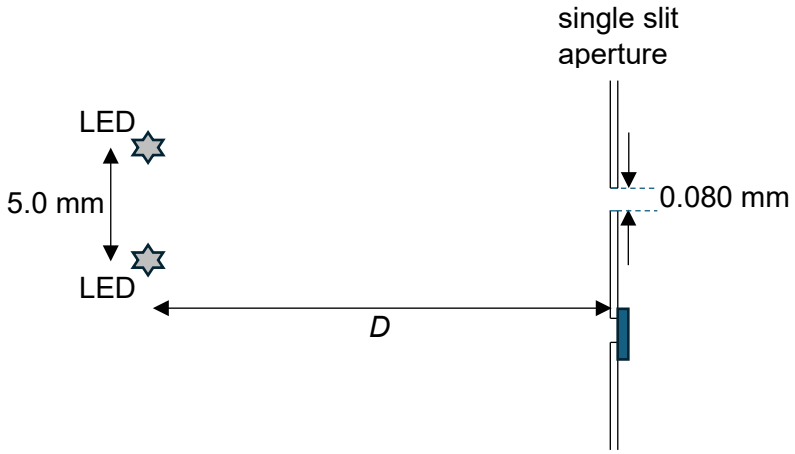


Figure 10.4 (not to scale)

Calculate the maximum distance *D* such that the images from the LED indicators can be just resolved.

D = ... m [3]

(d) A parallel beam of light of wavelength 633 nm is incident normally onto a 6.00 m screen. A bright spot is observed at the centre of the screen.

A diffraction grating is inserted normally into the path of the light as shown in Figure 10.5. The grating has 500 lines per millimetre.

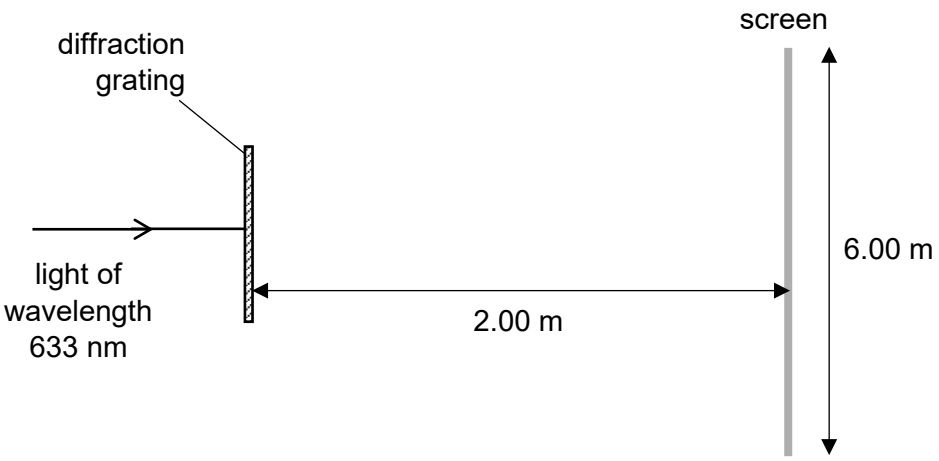


Figure 10.5 (not to scale)

(i) Determine the maximum number of bright lines that can be observed on the screen.

maximum number = …….. [4]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

(ii) The incident beam is now changed to a source of white light.

State and explain how the appearance of the first-order spectrum differs from the central maximum.

.....

.....

.....

..... [2]

[Total: 20]

~ End of Paper ~

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

4 **(a)** **(i)** State **one** difference and **one** similarity between the oscillations of a stationary sound wave and a progressive sound wave.

Difference: .....

.....

.....

Similarity: .....

.....

.....

[2]

(ii) Fig. 4.1 represents the displacement-time graph of two progressive sound waves travelling together through the same space.

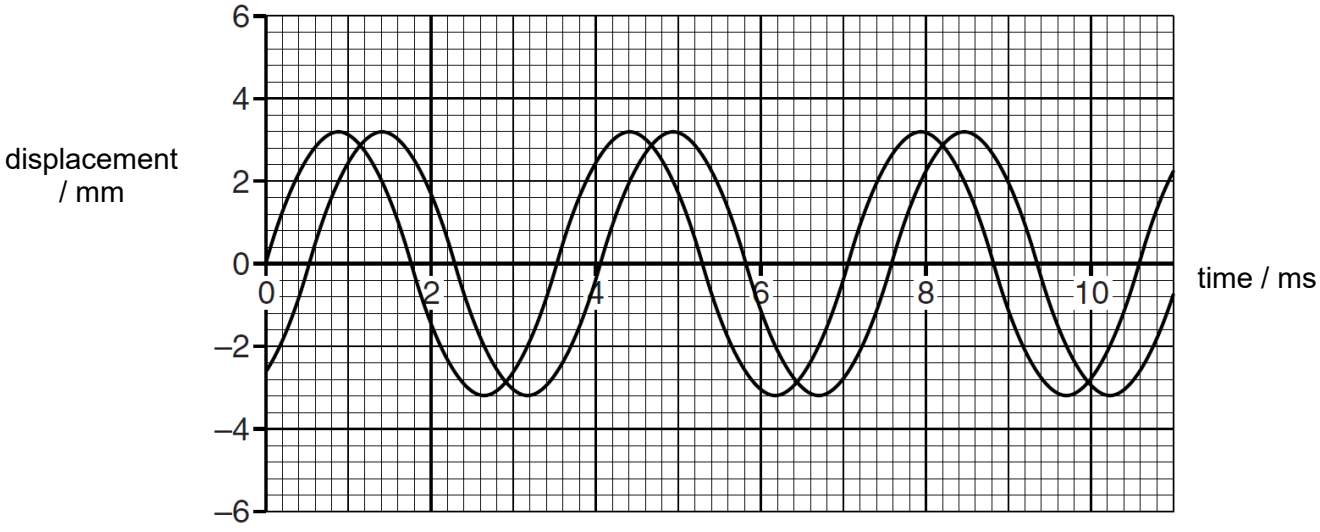


Fig. 4.1

1. State and explain whether Fig. 4.1 shows that the waves are coherent.

.....

.....

.....[1]

2. Use Fig. 4.1 to determine the phase difference between the waves.

phase difference = ... rad [1]

(iii) A sound wave travels along the axis of a tube that is open at both ends, as shown in Fig. 4.2. A stationary sound wave is formed in the tube.

The lowest frequency possible for the stationary sound wave in this tube is *f*.

On Fig. 4.2, mark all the positions of the displacement nodes (use the letter N) and the displacement antinodes (use the letter A) of a stationary sound wave for a frequency of 2*f*.

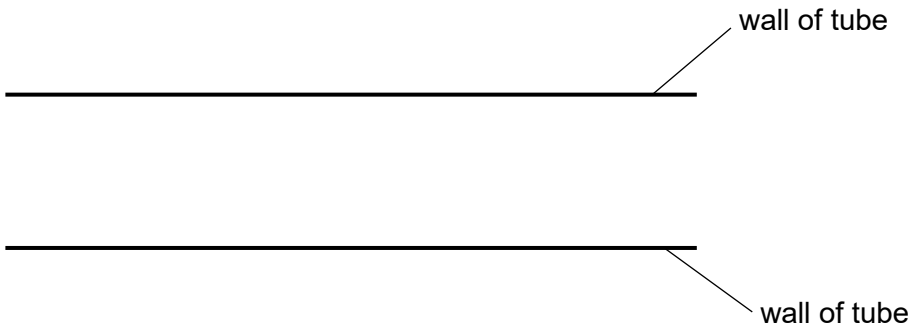


Fig. 4.2

[1]

(iv) With reference to the movement of air molecules on either side of a displacement node, explain why there is maximum variation in pressure at the nodes.

.....

.....

.....

.....

.....[2]

(b) Fig. 4.3 shows the arrangement for an experiment involving superposition of microwaves.

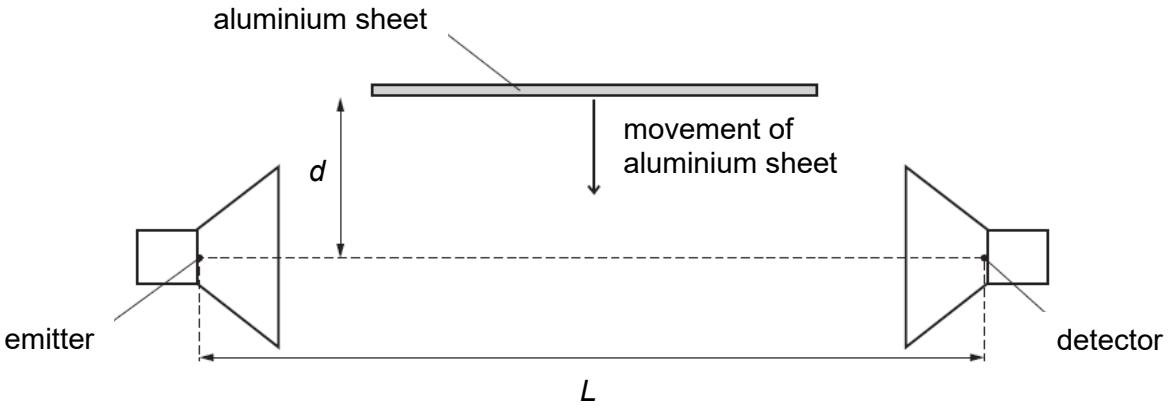


Fig. 4.3 (not to scale)

A microwave emitter and detector face each other with a distance *L* apart. A reflecting aluminium sheet is placed at a vertical distance *d* above the direct path between the emitter and the detector.

(i) When the aluminium sheet is moved in the direction shown by the arrow, the intensity of microwaves received by the detector fluctuates.

Describe and explain this variation in the microwave intensity.

.....

.....

.....

.....

.....[2]

(ii) As d decreases, the microwave detector has a maximum reading at d = 21.2 cm and the next maximum at d = 17.7 cm. The distance L stays constant at 80.0 cm.

Determine the wavelength λ of the microwaves.

λ = ... m [3]

(iii) Microwaves undergo a phase change of π rad when they reflect.

When d is much less than the wavelength of the microwaves, it is observed that the detector reading is a minimum.

Suggest why this observation demonstrates that microwaves must undergo a phase change on reflection from the aluminium sheet.

.....

.....

.....

.....[2]

[Total: 14]

3 A sound system in a large auditorium has two identical loudspeakers, S_1 and S_2, separated by 1.5 m. The amplitude of the sound emitted by each loudspeaker is proportional to the voltage across its terminals. A balance control is used to adjust the voltage supplied to each loudspeaker. The arrangement is shown in Fig. 3.1.

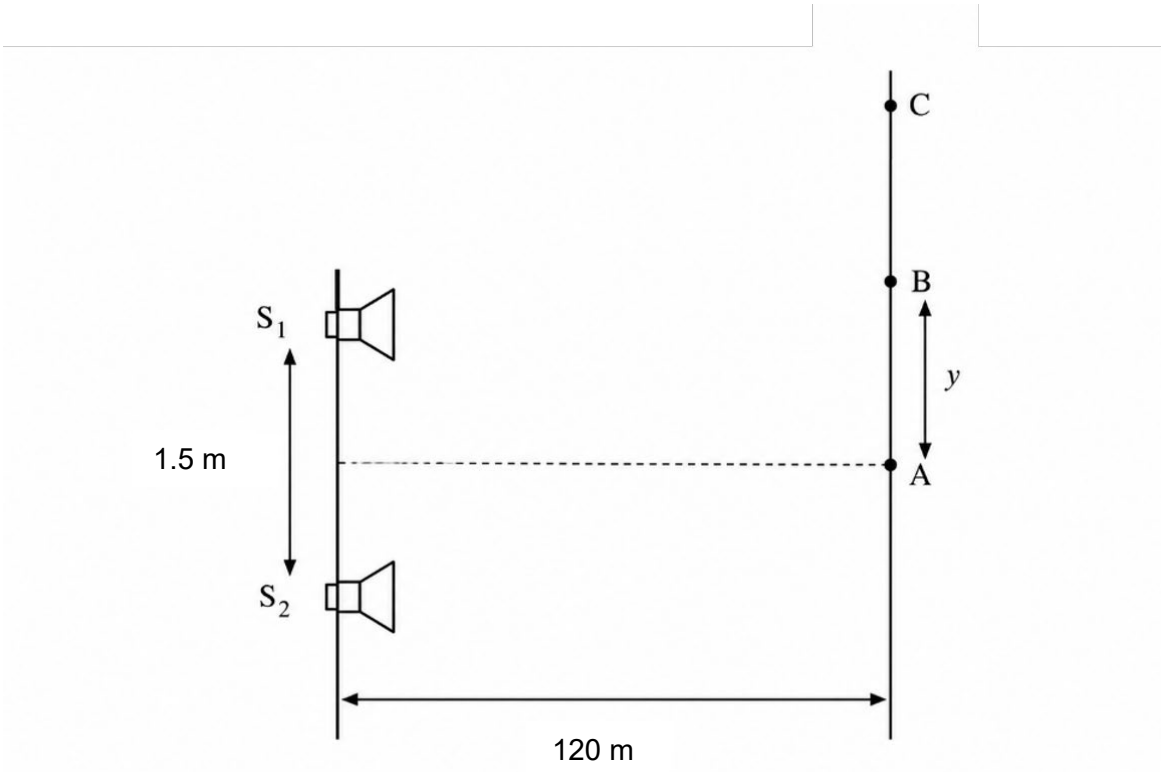


Fig. 3.1 (not to scale)

Initially, the loudspeakers emit sound waves of frequency 8500 Hz in phase. The balance control is adjusted so that the potential difference across each loudspeaker is 5.0 V r.m.s.

An observer at A hears a loud sound of intensity I_{max}. As he moves along the line AC, which is 120 m from the loudspeakers, the intensity first falls to zero at B, a distance *y* from A.

The speed of sound in air is 340 m s^{-1}.

(a) Determine the distance *y*.

y = ... m [3]

(b) Determine the next higher frequency of sound emitted by the speaker such that point B would also be a point of zero intensity.

frequency = ... Hz [3]

(c) The loudspeakers continue to emit sound of frequency 8500 Hz. The balance control is adjusted so that the voltages applied across S_1 and S_2 are 2.0 V r.m.s. and 8.0 V r.m.s. respectively. The amplitude of the sound emitted by each loudspeaker is proportional to the voltage applied across it.

In terms of I_{max}, determine the new intensity at B.

Intensity = ... I_{max} [3]

Total: [9]